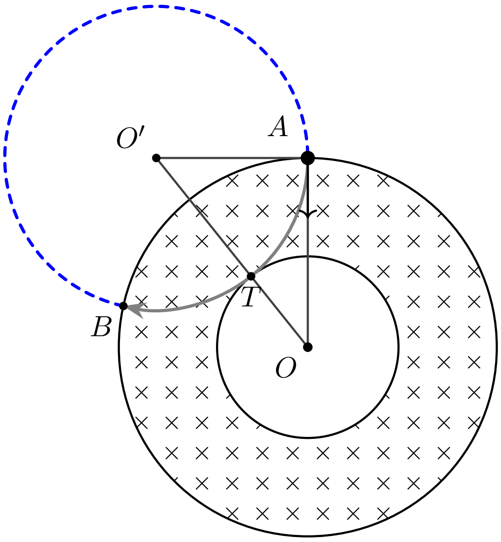

Q1. Magnetic Field

The particle experiences the Lorentz force

$$\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}.$$

Taking the entry point on the $+y$-axis, the inward velocity is $\mathbf{v} = -v\,\hat{\mathbf{j}}$ while $\mathbf{B} = B\,\hat{\mathbf{k}}$, so

$$\mathbf{F} = q\,(-v\hat{\mathbf{j}}) \times (B\hat{\mathbf{k}}) = -qvB\,(\hat{\mathbf{j}} \times \hat{\mathbf{k}}) = -qvB\,\hat{\mathbf{i}}.$$

For $q > 0$ the force points along $-\hat{\mathbf{i}}$, so the orbit bends to the left; for $q < 0$ it bends to the right. These are the two cases, and they give the same magnitude of B.

The force is purely centripetal, so the particle moves on a circle of gyroradius ρ with

$$\frac{mv^2}{\rho} = |q|\,vB \qquad \Longrightarrow \qquad \rho = \frac{mv}{|q|\,B}.$$

Let O' be the centre of this circle and T its point of tangency with the inner circle $r = R_1$. The velocity at A is radial, so $O'A \perp OA$ and the triangle $O\,A\,O'$ is right-angled at A, with

$$OA = R_2, \qquad AO' = \rho, \qquad OO' = \rho + R_1,$$

the last because OO' is the sum of the two touching radii ρ and R_1. Pythagoras then gives

$$(\rho + R_1)^2 = R_2^2 + \rho^2 \implies 2\rho R_1 = R_2^2 - R_1^2 \implies \rho = \frac{R_2^2 - R_1^2}{2R_1}.$$

Equating the two expressions for ρ and solving for B,

$$\frac{mv}{|q|\,B} = \frac{R_2^2 - R_1^2}{2R_1} \implies \boxed{B = \frac{2mvR_1}{|q|\,(R_2^2 - R_1^2)}}.$$